Lattice Yang–Mills theory at strong coupling IV: the mass gap is $4\log(1/\beta) + O(1)$

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

9 November 2026; revised after an independent referee report, and the matching upper bound added (Section 5), the same day; living document

Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d \geq 3$ and $0 < \beta < \beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II, and the finite-volume gaps of [F] on spatial lattices with all sides ≥ 4, are at least $4\log(1/q(\beta))$, with $q(\beta) = e^3 K(e^{4\beta} - 1)$ and $K = 4(2d-3)$. As $\beta \downarrow 0$ this is $4\log(1/\beta) - O(1)$. The leading coefficient 4 is the one predicted by the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, with predicted mass $-4\log u(\beta) + o(1)$, $u(\beta) \sim \beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta) - O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four (sides ≥ 4 are needed: with a side of length 3 the girth is three). In $d = 2$ the girth is the spatial size, which gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem. We also prove the matching upper bound: for an irreducible ρ of dimension N, $m \leq 4\log(1/u(\beta)) + O(\beta)$ with $u = c_\rho/(Nc_0)$, by an exact evaluation of the leading term of a plaquette–plaquette correlation (the four sides of a unit cube), a proof that every other term of the duplicated expansion has at least five plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\mathrm{Var}$ of the ground-state chain of Part II. Together: $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$, which is what principle 3 ("the strong-coupling test") of the research program requires. The upper bound is proved only for extremely small β (below about 10^{-16} for $SU(2)$ in $d = 4$), an artefact of sup-norm counting; its content is the asymptotics.

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1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient for all $\beta < \beta_0(d)$ (Theorem 1.2) and, for sufficiently small β, the matching upper bound (Theorem 1.3); together, $m(\mu_\beta) = 4\log(1/\beta) + O(1)$.

Literature. The strong-coupling glueball spectrum was studied rigorously in the 1980s: by their abstracts, Schor [Sch83] proved the existence of glueballs in strongly coupled lattice gauge theories, and O'Carroll, Braga and Schor [OBS85] (in $2+1$ dimensions) and O'Carroll and Barbosa [OB85] (in $3+1$ dimensions) obtained isolated masses of the form $-4\log\beta + r(\beta)$. We have not yet read these papers (they are marked "verify"), and nothing below relies on them. What this article adds for the project is a short self-contained proof with explicit constants and range, for the infinite-volume theory of [II], and a covariance bound for all bounded local observables.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2)},$$

where $c(\beta) := (e^{2\theta(\beta)}(1 + \frac{1}{K}))^{2(d-1)}$.

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Theorem 1.3 (Upper bound; the gap is $4\log(1/\beta)+O(1)$)**.** *Let G,ρ be as in Theorem 1.2 (G metrisable), $d\geq 3$, and let ρ be irreducible and non-trivial (of dimension N); put $\iota:=1$ if $\rho\cong\bar\rho$ and $\iota:=0$ otherwise, and $u(\beta):=c_\rho(\beta)/(Nc_0(\beta))$. There are explicit functions $\delta(\beta),D(\beta)=O(\beta)$ (Theorem 5.4) such that, whenever $0<\beta<\beta_0(d)$ and $\delta(\beta)<e^{-48(d-1)\theta(\beta)}$,*

$$m(\mu_\beta)\leq 4\log\frac{1}{u(\beta)}+D(\beta),$$

and the same bound holds for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 3. Consequently, as $\beta\downarrow 0$,

$$-4\log(4e^3K)+O(\beta)\ \leq\ m(\mu_\beta)-4\log\frac{1}{\beta}\ \leq\ 4\log\frac{2N^2}{1+\iota}+O(\beta),$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the spatial lattices with all sides ≥ 4.

Here $\mathrm{Cov}(A,B):=\mu(AB)-\mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) follows from Theorem 4.1 by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)], and (ii) from (iii) by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii).

Remark 1.4 (Comparison). For $d=4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta)-2.89+O(\beta^2)$; the bound here is $4\log(1/\beta)-29.5+O(\beta)$. In the common range of validity $\beta<\beta_0(4)\approx 1.54\cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (5.890 and 5.591 at $\beta_0(4)$), and the present one is larger for $\beta<1.39\cdot 10^{-4}$, by an additive amount that tends to infinity as $\beta\to 0$ (the ratio of the two bounds increases to 4; it is 1.92 at $\beta=10^{-5}$). These values were computed with `python3`. The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant -29.5 is far from the predicted $+4\log 4$ for $SU(2)$ (from $u\approx\beta/4$).

Strategy. (1) Let U and U' be two independent copies of the field, each with law μ. Then $2\,\mathrm{Cov}_\mu(A,B)$ is the integral of $(A-A')(B-B')$ against $\mu\otimes\mu$. The duplicated Wilson weight is $\prod_p(1+f_\beta(U_p))(1+f_\beta(U'_p))=\prod_p(1+g_p)$ with $|g_p|\leq e^{4\beta}-1$, so the loop expansion of [III] applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U\leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.2), hence at least $4\times$(number of layers) plaquettes (Corollary 3.4). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III]. (5) For the upper bound (Section 5) the same expansion is applied to two plaquettes at times 0 and 1: the four sides of the unit cube between them form the unique minimal connecting set, its term in the duplicated expansion is computed exactly by character orthogonality (up to a ratio of partition functions equal to $1+O(\beta)$), all other terms have at least five plaquettes, and the spectral inequality $\mathrm{Cov}\leq e^{-m}\,\mathrm{Var}$ converts the resulting lower bound on the covariance into an upper bound on m.

2 The duplicated system

Throughout this section and the next, $\Lambda=\prod_{\lambda=0}^{d-1}\mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda\geq 3$, with the conventions of [F, III]: $\mathcal{P}=\mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta=w_\beta/c_0(\beta)-1$, and for $\mathcal{Q}\subseteq\mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q})=\int\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\,dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P})=Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\,dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U,U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p\in X} g_p \, dU\, dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \le \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p\in\mathcal{Q}}(1+g_p)\, dU\, dU' = \Xi(\mathcal{Q})^2$*; and* $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ *has density* $\prod_{p\in\mathcal{P}}(1+g_p)/\Xi(\mathcal{P})^2$ *with respect to* $dU\,dU'$.

(c) Let $X \subseteq \mathcal{P}$*, let* ℓ *be a link lying in exactly one plaquette of* X*, and let* H *be bounded measurable and independent of* (U_ℓ, U'_ℓ)*. Then* $\iint H \prod_{p\in X} g_p\, dU\, dU' = 0$.

Proof. (a) By [III, Lemma `al:lem:facts`](a),(b), $e^{-\beta} \le w_\beta \le e^\beta$ and $1 \le c_0 \le e^\beta$, so $1+f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^\beta]$ and $(1+f_\beta(U_p))(1+f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta}-1, e^{2\beta}-1]$; both $1-e^{-4\beta}$ and $e^{2\beta}-1$ are at most $e^{4\beta}-1$. (b) The integrand is $\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\cdot\prod_{p\in\mathcal{Q}}(1+f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U)W_\beta(U') = c_0^{2|\mathcal{P}|}\prod_p(1+g_p)$ and $Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm1}B$ and $U'_p = A'U'^{\pm1}_\ell B'$ with A, B, A', B' independent of (U_ℓ, U'_ℓ), and no other factor of the integrand depends on (U_ℓ, U'_ℓ). By Fubini it suffices that $\iint g_p\, dU_\ell\, dU'_\ell = \int(1 + f_\beta(AU^{\pm1}B))\, dU \cdot \int(1 + f_\beta(A'U^{\pm1}B'))\, dU - 1 = 1\cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p) \cap S \ne \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X) \cup S) \ne \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \le J_S$ and $|\overline{X}| \le (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let* $S \subseteq E(\Lambda)$ *and let* F *be bounded, measurable, and dependent only on the variables* (U_ℓ, U'_ℓ)*,* $\ell \in S$*. Then*

$$\int F\, d(\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}) = \sum_{X\in\mathcal{X}_S} \Phi_F(X)\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U'_ℓ), dU by $dU\,dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\, d(\mu\otimes\mu) = \sum_{T\subseteq\mathcal{P}} \iint F \prod_{p\in T} g_p\, dU\, dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P}\setminus\overline{X}$. For such a pair, $F\prod_{p\in X} g_p$ depends only on the variables of the links in $S \cup E(X)$ and $\prod_{p\in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y\subseteq\mathcal{P}\setminus\overline{X}} \iint \prod_{p\in Y} g_p = \iint \prod_{p\in\mathcal{P}\setminus\overline{X}}(1+g_p) = \Xi(\mathcal{P}\setminus\overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let* A, B *be bounded measurable functions on* Ω_Λ *depending only on disjoint link sets* S_A, S_B*, put* $S := S_A \cup S_B$ *and* $F(U,U') := (A(U)-A(U'))(B(U)-B(U'))$*. Then:*

(i) $\int F\, d(\mu\otimes\mu) = 2\,\mathrm{Cov}_\mu(A,B)$ *for every probability measure* μ *on* Ω_Λ*, and* $\|F\|_\infty \le 4\|A\|_\infty\|B\|_\infty$*;*

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X) \setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A - A')(B - B')\, d(\mu \otimes \mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X \setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1) \cap S_B = \emptyset$, and likewise $E(X_2) \cap S_A = \emptyset$; and $E(X_1) \cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A \cup E(X_1)$ and $S_B \cup E(X_2)$ are disjoint, and since $dU\, dU'$ is a product measure,

$$\Phi_F(X) = \iint (A - A') \prod_{p \in X_1} g_p \cdot \iint (B - B') \prod_{p \in X_2} g_p.$$

The exchange $(U, U') \mapsto (U', U)$ preserves $dU\, dU'$ and every g_p (which is symmetric in U_p, U'_p), and changes the sign of $A - A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F \prod_{p' \in X \setminus \{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x, 0)$ is *temporal of layer* x_0 (it joins times x_0 and $x_0 + 1$); a plaquette $(x, 0i)$, $1 \le i \le d-1$, is *temporal of layer* x_0. For $Y \subseteq \mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one). Let $\Lambda_s := \prod_{i=1}^{d-1} \mathbb{Z}/L_i\mathbb{Z}$ be the spatial torus and g the *girth* of its nearest-neighbour graph (the length of a shortest cycle).

Lemma 3.1 (Girth)**.** *Let all $L_i \ge 3$. If $d \ge 3$, then $g = 4$ when all $L_i \ge 4$, and $g = 3$ when some $L_i = 3$. If $d = 2$, then $g = L_1$.*

Proof. Since all $L_i \ge 3$ the graph is simple. A cycle is a closed walk without backtracking, i.e. a sum of unit vectors $\pm e_i$ equal to 0 in Λ_s with no two consecutive steps opposite. For $d = 2$ the graph is the cycle $\mathbb{Z}/L_1\mathbb{Z}$ itself. For $d \ge 3$ a unit square is a cycle of length 4; a cycle of length 3 needs $\pm e_i \pm e_j \pm e_k = 0$ with no two opposite consecutive steps, which forces $i = j = k$ and $3e_i = 0$, i.e. $L_i = 3$; and conversely $L_i = 3$ gives the cycle $0, e_i, 2e_i$. □

Lemma 3.2 (Plaquettes per layer)**.** *Let all $L_i \ge 3$ $(1 \le i \le d-1)$. Let $S \subseteq E(\Lambda)$ and $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \neq \emptyset$. Then $|Y_s| \ge g$. In particular $|Y_s| \ge 4$ if $d \ge 3$ and all $L_i \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s, \vec{x}), 0)$, $\vec{x} \in \Lambda_s$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s, \vec{x}), 0i)$ and $((s, \vec{x} - e_i), 0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s, \vec{x}), 0i) \in Y_s$, each joining $\vec{x}$ and $\vec{x} + e_i$ (the positions of its two temporal links). Since $L_i \ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a subgraph of the nearest-neighbour graph of Λ_s. If $\vec{x}$ is an endpoint of an edge, then $\ell_{\vec{x}} \in E(Y) \setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \neq \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats), which has at least g edges; they are distinct plaquettes of Y_s. The last statement is Lemma 3.1. □

Lemma 3.3 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \neq s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1, \dots, s'\}$ and $I_2 := \{s'+1, \dots, s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s, s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.4 (Size of connecting components)**.** *Let all $L_\lambda \geq 3$. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \leq a_2 < b_1 \leq b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1, \dots, a_2\}$ and those of S_B in $\{b_1, \dots, b_2\}$ (modulo L_0). Put*

$$\tau_{\text{in}} := b_1 - a_2, \qquad \tau_{\text{out}} := L_0 - (b_2 - a_1), \qquad \tau := \min(\tau_{\text{in}}, \tau_{\text{out}}) \ \ (\geq 1).$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq g\,\tau$. More precisely, with the inner layers $\{a_2, \dots, b_1 - 1\}$ and the outer layers $\{b_2, \dots, a_1 - 1 + L_0\}$, Y contains at least g temporal plaquettes in every inner layer, or at least g temporal plaquettes in every outer layer.

Proof. Call the layers $s \in \{a_2, \dots, b_1 - 1\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2, \dots, a_1 - 1 + L_0\}$ outer (τ_{out} of them); as integers, $s < s' < s + L_0$ for every inner s and outer s', so inner and outer layers are distinct. *These layers are free.* A temporal link of an inner layer s joins s and $s+1$, where $s+1 \in \{a_2+1, \dots, b_1\}$ is not in $\{a_1, \dots, a_2\}$ and $s \in \{a_2, \dots, b_1 - 1\}$ is not in $\{b_1, \dots, b_2\}$ modulo L_0 (because $b_2 - a_1 < L_0$); so it belongs neither to S_A nor to S_B. The same argument applies to outer layers. *Times of the plaquettes of Y at the supports.* Let $p \in Y \cap \mathcal{P}_{S_A}$, containing a link $\ell \in S_A$. If p is spatial, $T(p) = \{r\}$ with $r \in \{a_1, \dots, a_2\}$; if p is temporal of layer r, then ℓ is either its temporal link of layer r or a spatial link at time r or $r+1$, so $r \in \{a_1 - 1, \dots, a_2\}$. Now let s be inner and s' outer with $Y_s = Y_{s'} = \emptyset$, and I_1, I_2 as in Lemma 3.3. Then $T(p) \subseteq I_2 = \{s'+1, \dots, s\}$: the times $a_1, \dots, a_2$ lie in I_2 because $s' + 1 \leq a_1 + L_0$ and $s \geq a_2$; the time $a_2 + 1$ occurs in $T(p)$ only if p is temporal of layer a_2, which is excluded when $s = a_2$ (as $Y_s = \emptyset$) and lies in I_2 when $s \geq a_2 + 1$; similarly the time $a_1 - 1$ occurs only if p is temporal of layer $a_1 - 1$, which is excluded when $s' \equiv a_1 - 1$ and lies in I_2 otherwise. In the same way $T(p) \subseteq I_1 = \{s+1, \dots, s'\}$ for every $p \in Y \cap \mathcal{P}_{S_B}$. *Conclusion.* If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.3 puts all of Y into one of the disjoint classes I_1, I_2, which contradicts the previous paragraph, since Y meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.2 (with $S = S_A \cup S_B$) gives at least g plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \geq 2$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \geq 3$; let g be the girth of Lemma 3.1. Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.4, with $\tau = \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta) < 1$ (Lemma 4.2) and $c = c(\beta)$ as in Theorem 1.2,*

$$\left|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)\right| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q}\, c^{|S_A| + |S_B|}\, q^{g\tau}.$$

Proof. If S_A or S_B is empty, A or B is constant and the covariance vanishes; so let both be non-empty. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\operatorname{Cov}(A,B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X)\setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.4 these have $|X| \geq n_0 := g\tau$. For each of them: $|\Phi_F(X)| \leq 4\|A\|_\infty\|B\|_\infty\,\epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2 \leq e^{2\theta|\overline{X}|} \leq e^{2\theta(K+1)|X|}e^{2\theta J_S} \leq e^{2|X|}e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S+Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \leq J_S$ and the adjacency bound K; it requires $J_S \geq 1$, i.e. $S \neq \emptyset$). With $\binom{m}{n}t^n \leq (1+t)^m$ at $t = 1/K$, $\binom{J_S+Kn}{n} \leq (1+\frac{1}{K})^{J_S}(1+\frac{1}{K})^{Kn}K^n \leq (1+\frac{1}{K})^{J_S}(eK)^n$. Altogether

$$2|\operatorname{Cov}(A,B)| \leq 4\|A\|_\infty\|B\|_\infty\, e^{2\theta J_S}\Big(1+\frac{1}{K}\Big)^{J_S}\sum_{n\geq n_0}\big(eK\cdot e^2\epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-q}\,q^{n_0},$$

since $e^3K\epsilon(2\beta) = q$ and $(e^{2\theta}(1+\frac{1}{K}))^{J_S} = c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d \geq 2$ and $0 < \beta < \beta_0(d)$: $q(\beta) < \frac{e(2+\epsilon_0)}{K+2}$, which is < 0.909 for $d = 2$ and < 0.389 for $d \geq 3$; and $4\log\frac{1}{q(\beta)} = 4\log\frac{1}{\beta} - 4\log(4e^3K) + O(\beta)$ as $\beta \downarrow 0$. Moreover $\beta_0(d) < \beta_*(d) = \operatorname{artanh}\frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta}-1 < \epsilon_0$ (which is the condition $\beta < \beta_0(d)$), $e^{4\beta}-1 = \epsilon(2+\epsilon) < \epsilon_0(2+\epsilon_0)$ and $e^3K\epsilon_0 = \frac{e}{K+2}$; for $d = 2$, $K = 4$ and $\frac{e(2+\epsilon_0)}{6} < 0.909$; for $d \geq 3$, $K \geq 12$ and $\frac{e(2+\epsilon_0)}{14} < 0.389$. The asymptotics follow from $e^{4\beta}-1 = 4\beta+O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta < \beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde{\mu}_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A \cup S_B$, the functions $A\circ\iota$, $B\circ\iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i, b_i, $g = 4$, $\tau_{\text{in}} = b_1 - a_2$ and $\tau_{\text{out}} \to \infty$. Theorem 4.1 bounds $\operatorname{Cov}_{\tilde{\mu}_{\Lambda_k,\beta}}(A,B)$ uniformly in k, and the left side converges to $\operatorname{Cov}_{\mu_\beta}(A,B)$.

(ii) Let $F \in \mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0, n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F, 0]$ and S_tF on links with endpoint times in $[t, t+n_F]$, with the same number of links. For $t \geq 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_tF) = |\mu_\beta(F)|^2$),

$$\big|\mu_\beta(\Theta F\cdot S_tF) - |\mu_\beta(F)|^2\big| = \big|\operatorname{Cov}_{\mu_\beta}(\Theta F, S_tF)\big| \leq C_F\, q^{4t} \qquad (t \geq 1),$$

with C_F independent of t (here $b_1 - a_2 = t$); for $t = 0$ the left side is at most $2\|F\|_\infty^2$. So the hypothesis of [II, Theorem `iv:thm:OS`](iii) holds with $m = 4\log(1/q)$ and the constant $\max(C_F, 2\|F\|_\infty^2)$, and $m(\mu_\beta) \geq 4\log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s = G^{E(\Lambda_s)}$ and $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \geq 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.4 take $a_1 = a_2 = 0$, $b_1 = b_2 = t$; for $1 \leq t \leq L_0/2$, $\tau = \min(t, L_0 - t) = t$, and $g = 4$ since all $L_i \geq 4$. Theorem 4.1 gives $|\langle \overline{A}(t) A(0)\rangle - \langle \overline{A}(t)\rangle\langle A(0)\rangle| \leq C_A q^{4t}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0; for $t = 0$ the left side is at most $2\|A\|_\infty^2$. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0 \to \infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta) \geq 4\log(1/q)$. □

5 The matching upper bound

Throughout this section $d \geq 3$, G is metrisable (as in Theorem 1.2), and ρ is irreducible and non-trivial (automatic for an irreducible faithful ρ when $G \neq \{1\}$); $N = d_\rho$, $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, and $u(\beta) := c_\rho(\beta)/(N c_0(\beta))$. Fix two spatial directions, labelled $1, 2$. On a periodic lattice Λ (all $L_\lambda \geq 3$) let $p_0 := (0, 12)$ be the spatial plaquette at the origin and time 0, $p_1 := (e_0, 12)$ its translate at time 1, and

$$A := \operatorname{Re}\chi_\rho(U_{p_0}), \qquad A_1 := \operatorname{Re}\chi_\rho(U_{p_1}), \qquad S_A := E(p_0), \quad S_B := E(p_1), \quad S := S_A \cup S_B,$$

$F(U, U') := (A(U) - A(U'))(A_1(U) - A_1(U'))$ as in Lemma 2.5, and let X_0 be the set of the four temporal plaquettes $(y, 0i)$ of layer 0 with $(y, i) \in E(p_0)$: the sides of the unit cube Q with bottom p_0 and top p_1. The same notation is used on $\mathbb{Z}^d$.

Lemma 5.1 (Leading terms)**.** *For every $\beta > 0$:*

(i) $\tau_0 := \int A^2\, dU = \frac{1+\iota}{2}$;

(ii) $\tau_1 := \int A\, A_1 \prod_{s\in X_0} f_\beta(U_s)\, dU = \tau_0\, u(\beta)^4$;

(iii) $\Phi_F(X_0) = 2\tau_1$;

(iv) $\frac{(1+\iota)\beta e^{-\beta}}{2N^2} \leq u(\beta) \leq \frac{(1+\iota)\beta + \beta^2 e^\beta}{2N^2}$.

Proof. (i) With $\operatorname{Re}\chi = \frac{1}{2}(\chi + \bar{\chi})$, $\int (\operatorname{Re}\chi_\rho)^2 = \frac{1}{4}(\int \chi_\rho^2 + 2\int |\chi_\rho|^2 + \int \overline{\chi_\rho}^2)$. By Schur orthogonality [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii), $\int |\chi_\rho|^2 = 1$ and, since $\chi_{\bar\rho} = \overline{\chi_\rho}$, $\int \chi_\rho^2 = \int \chi_\rho \overline{\chi_{\bar\rho}} = \iota$; likewise $\int \overline{\chi_\rho}^2 = \iota$.

(ii) Orient the six faces of Q as its boundary. The proof of [F, Proposition 'Cube activity' (`prop:cube`)] establishes, in its line "Altogether $\int \prod_p \chi_{r_p}(U_p)\, dU = \delta_{\text{all } r_p = r}\, d_r^{-4}$" (whose derivation does not use $r_p \neq 0$): for irreducible representations attached to the six faces, $\int \prod_{\text{faces}} \chi_{r_{\text{face}}}(U_{\text{face}})\, dU$ equals d_r^{-4} if all six are equivalent to the same r, and 0 otherwise. Each boundary-oriented face holonomy is conjugate to U_p or U_p^{-1}, and f_β and $\operatorname{Re}\chi_\rho$ are class functions invariant under inversion, so A, A_1 and $f_\beta(U_s)$ are the same functions of the boundary-oriented holonomies. Insert $f_\beta = \sum_{r\neq 0}(c_r/c_0)\chi_r$ (absolutely and uniformly convergent, [F, Corollary `cor:uniform`]) for the four sides, and $A = \chi_\rho$ if $\iota = 1$ (χ_ρ is then real), $A = \frac{1}{2}(\chi_\rho + \chi_{\bar\rho})$ if $\iota = 0$, and the same for A_1. Only the terms in which all six faces carry the same $r \in \{\rho, \bar\rho\}$ survive. As $c_{\bar\rho} = \int \chi_\rho w_\beta\, dU = \overline{c_\rho}$ (w_β is real), $c_\rho > 0$ ([F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]), and $d_{\bar\rho} = d_\rho = N$, this gives $\tau_1 = (c_\rho/c_0)^4 N^{-4} = u^4$ if $\iota = 1$ and $\tau_1 = \frac{1}{4}\cdot 2\cdot (c_\rho/c_0)^4 N^{-4} = \frac{1}{2}u^4$ if $\iota = 0$; in both cases $\tau_1 = \tau_0 u^4$.

(iii) Expand $\prod_{s\in X_0} g_s = \prod_s (f_s + f_s' + f_s f_s')$, where $f_s := f_\beta(U_s)$ and $f_s' := f_\beta(U_s')$, and for each of the 3^4 terms let $Q_1 \subseteq X_0$ ($Q_2 \subseteq X_0$) be the set of sides carrying a factor f_s (a factor f_s'); $Q_1 \cup Q_2 = X_0$. Each of the four temporal links of Q lies in exactly two sides, adjacent around the square, and in no other factor of the integrand. If some temporal link lies in exactly one plaquette of Q_1, integrating it out in the first copy gives 0 ([III, Lemma `al:lem:facts`](c)); so $Q_1 \in \{\emptyset, X_0\}$, and likewise $Q_2 \in \{\emptyset, X_0\}$. Three terms remain: $\prod f_s$,

$\prod f'_s$ and $\prod f_s f'_s$. Expand $F = AA_1 - AA'_1 - A'A_1 + A'A'_1$; the integral over $dU\,dU'$ factorises into the two copies. A factor $\int A \prod_s f_s\,dU$, $\int A_1 \prod_s f_s\,dU$ or $\int \prod_s f_s\,dU$ vanishes, because the top (resp. bottom) edges of Q then lie in exactly one factor; the same holds in the second copy. Hence the term $\prod f_s$ contributes $\int AA_1 \prod_s f_s\,dU = \tau_1$, the term $\prod f'_s$ contributes τ_1, and the term $\prod f_s f'_s$ contributes 0.

(iv) By the proof of [F, Lemma `lem:cr`], $c_\rho = \sum_{n,m\geq 0} \frac{t^{n+m}}{n!\,m!} m_\rho(n,m)$ with $t = \beta/(2N)$ and $m_\rho(n,m)$ the multiplicity of ρ in $\rho^{\otimes n} \otimes \bar{\rho}^{\otimes m}$. Here $m_\rho(0,0) = 0$, $m_\rho(1,0) = 1$, $m_\rho(0,1) = \iota$, and $m_\rho(n,m) \leq N^{n+m-1}$ (dimension count). Hence $(1+\iota)t \leq c_\rho \leq (1+\iota)t + N^{-1}(e^{2Nt} - 1 - 2Nt) \leq \frac{(1+\iota)\beta}{2N} + \frac{\beta^2 e^\beta}{2N}$, using $e^y - 1 - y \leq \frac{y^2}{2} e^y$ with $y = 2Nt = \beta$. Divide by Nc_0 and use $1 \leq c_0 \leq e^\beta$ ([III, Lemma `al:lem:facts`](b)). □

Lemma 5.2 (Minimal connecting set)**.** *Let Λ be periodic with all $L_\lambda \geq 3$. Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq 4$, with equality only if $Y = X_0$. Consequently, $\Phi_F(X) = 0$ for every $X \in \mathcal{X}_S$ with $|X| \leq 4$ and $X \neq X_0$.*

Proof. Apply Corollary 3.4 with $a_1 = a_2 = 0$, $b_1 = b_2 = 1$: the only inner layer is 0, the outer layers are $1, \dots, L_0 - 1$ (at least two), and $g \geq 3$ (Lemma 3.1). If Y has temporal plaquettes in every outer layer, $|Y| \geq 3 \cdot 2 > 4$. Otherwise $|Y_0| \geq g \geq 3$. Suppose $|Y| \leq 4$. The temporal plaquette $(y, 0i)$ of layer 0 contains exactly one spatial link at time 0, its bottom edge (y, i), and distinct such plaquettes have distinct bottom edges. A bottom edge of a plaquette of Y_0 lies in no plaquette of another component; it is not in S_B (whose links are at time 1); so, unless it lies in S_A, it lies in a second plaquette of Y, not in Y_0.

If $|Y_0| = 4$, then $Y = Y_0$, so all four bottom edges lie in $S_A = E(p_0)$; being four distinct links, they are the four edges of p_0, and $Y = X_0$.

If $|Y_0| = 3$, the graph Γ of the proof of Lemma 3.2 has three edges and minimum degree 2 at its non-isolated vertices, so it is a triangle; by Lemma 3.1 some $L_i = 3$ and the bottom edges are the three collinear links $(y, i), (y + e_i, i), (y + 2e_i, i)$ at time 0. At most one further plaquette p' belongs to Y, and p' contains at most one of them (the links of a plaquette in direction i are (z, i) and $(z + e_j, i)$ with $j \neq i$, and a temporal plaquette contains one spatial link at time 0). So at least two of the three lie in $S_A = E(p_0)$, which contains at most one link of a collinear triple: if $i \in \{1, 2\}$, its links in direction i are $(0, i)$ and (e_j, i), $\{i, j\} = \{1, 2\}$, which are not collinear; if $i \geq 3$, it contains no link in direction i. This contradiction shows $|Y_0| \neq 3$.

For the last statement: if $\Phi_F(X) \neq 0$, Lemma 2.5 gives no free edges outside S and a component Y as above, so $|X| \geq |Y| \geq 4$, and $|X| = 4$ forces $X = Y = X_0$. □

Lemma 5.3 (Two-point bounds in finite volume)**.** *Let $0 < \beta < \beta_0(d)$ and let Λ be periodic with all $L_\lambda \geq 3$. Put*

$$\hat{q}(\beta) := eK\, e^{2(K+1)\theta(\beta)} \big(e^{4\beta} - 1\big) \;\leq\; q(\beta) < 1, \qquad c(\beta) = \big(e^{2\theta(\beta)}(1 + \tfrac{1}{K})\big)^{2(d-1)}.$$

Then

$$\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, A_1) \geq \tau_1\, e^{-48(d-1)\theta} - \frac{2N^2 c^8 \hat{q}^{\,5}}{1 - \hat{q}},$$
$$\mathrm{Var}_{\mu_{\Lambda,\beta}}(A) \leq \tau_0\, e^{8(d-1)\theta} + N^2 \big(e^\theta(1 + \tfrac{1}{K})\big)^{8(d-1)} \frac{x}{1 - x}.$$

Proof. Covariance. By Lemmas 2.4 and 2.5(i), $2\,\mathrm{Cov}(A, A_1) = \sum_{X \in \mathcal{X}_S} \Phi_F(X) r_X^2$ with $r_X := \Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P})$. The term X_0 equals $2\tau_1 r_{X_0}^2$ (Lemma 5.1(iii)), and $r_{X_0}^2 \geq e^{-2\theta|\overline{X_0}|}$ by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)]; since $S \subseteq E(X_0)$ and $E(X_0)$ consists of the twelve edges of Q, $\overline{X_0}$ is the set of plaquettes containing one of these edges, and

$|\overline{X_0}| \le 24(d-1)$. By Lemma 5.2 every other non-zero term has $|X| \ge 5$. Bound them as in the proof of Theorem 4.1, but keeping the factor $e^{2\theta(K+1)|X|}$ instead of $e^{2|X|}$: with $J_S = 16(d-1)$ and $\|A\|_\infty, \|A_1\|_\infty \le N$,

$$\sum_{X \neq X_0} |\Phi_F(X)| r_X^2 \le 4N^2 e^{2\theta J_S} \Big(1 + \frac{1}{K}\Big)^{J_S} \sum_{n \ge 5} \big(eK\, e^{2\theta(K+1)} \epsilon(2\beta)\big)^n = \frac{4N^2 c^8 \hat{q}^{\,5}}{1-\hat{q}}.$$

Divide by 2.

Variance. $\mathrm{Var}(A) \le \langle A^2 \rangle$. The proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] applies verbatim to the function A^2, which depends only on the links of S_A, with E_C replaced by S_A; it gives $\langle A^2 \rangle = \sum_{X \in \mathcal{X}_{S_A}} \Phi_{A^2}(X)\, \Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P})$ with $\Phi_{A^2}(X) := \int A^2 \prod_{p \in X} f_\beta(U_p)\, dU$. The term $X = \emptyset$ is $\tau_0\, \Xi(\mathcal{P} \setminus \mathcal{P}_{S_A})/\Xi(\mathcal{P}) \le \tau_0 e^{\theta |\mathcal{P}_{S_A}|} \le \tau_0 e^{8(d-1)\theta}$. For the others, $|\Phi_{A^2}(X)| \le N^2 \epsilon^{|X|}$, the ratio is at most $e^{\theta(K+1)|X| + \theta J_{S_A}}$ with $J_{S_A} = 8(d-1)$, and there are at most $(1+\frac{1}{K})^{J_{S_A}} (eK)^n$ sets with $|X| = n$ (as in the proof of Theorem 4.1); since $eK e^{\theta(K+1)} \epsilon \le e^2 K \epsilon = x$, their sum is at most $N^2 (e^\theta (1 + \frac{1}{K}))^{8(d-1)} \sum_{n \ge 1} x^n$. □

Theorem 5.4 (Upper bound)**.** *Let $d \ge 3$, G metrisable, ρ irreducible and non-trivial, $0 < \beta < \beta_0(d)$, and with the notation of Lemma 5.3 suppose that*

$$\delta(\beta) := \frac{4N^2}{1+\iota} \, \frac{c(\beta)^8\, \hat{q}(\beta)^5}{(1-\hat{q}(\beta))\, u(\beta)^4} \;<\; e^{-48(d-1)\theta(\beta)}.$$

Put

$$D(\beta) := \log \frac{e^{8(d-1)\theta} + \frac{2N^2}{1+\iota} \big(e^\theta (1 + \frac{1}{K})\big)^{8(d-1)} \frac{x}{1-x}}{e^{-48(d-1)\theta} - \delta(\beta)}.$$

Then

(i) $m(\mu_\beta) \le 4 \log \frac{1}{u(\beta)} + D(\beta)$;

(ii) $m_{\Lambda_s}(\beta) \le 4 \log \frac{1}{u(\beta)} + D(\beta)$ *for every periodic spatial lattice Λ_s with all sides ≥ 3.*

Moreover $\delta(\beta) = O(\beta)$ and $D(\beta) = O(\beta)$ as $\beta \downarrow 0$; in particular the hypothesis holds for all sufficiently small β.

Proof. Write Γ_1 and Γ_0 for the covariance of A_1 and A and for the variance of A in the relevant theory. By Lemma 5.3, $\tau_0/\tau_1 = u^{-4}$ (Lemma 5.1) and $2N^2/\tau_1 = \frac{4N^2}{(1+\iota)u^4}$,

$$\Gamma_1 \ge \tau_1 \big(e^{-48(d-1)\theta} - \delta(\beta)\big) > 0, \qquad \Gamma_0 \le \tau_0 \Big(e^{8(d-1)\theta} + \frac{2N^2}{1+\iota} \big(e^\theta (1 + \tfrac{1}{K})\big)^{8(d-1)} \frac{x}{1-x}\Big),$$

so $\log(\Gamma_0/\Gamma_1) \le 4 \log(1/u) + D(\beta)$, provided these bounds hold for the theory in question. It remains to show $m \le \log(\Gamma_0/\Gamma_1)$.

(ii) Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \ge 3$, and A the corresponding gauge-invariant function on the spatial configurations, so that $A_1 = A(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. By [F, Theorem 'Ground-state expectations' (`thm:gs`)] (applied to $A_1 A$, to A and to A^2, using $\langle A(1) \rangle_\Lambda = \langle A(0) \rangle_\Lambda$ by translation invariance, [F, Theorem `thm:basic`](iv)), as $L_0 \to \infty$, $\mathrm{Cov}_\Lambda(A_1, A) \to \langle Q\hat{A}\Omega, e^{-H} Q\hat{A}\Omega \rangle$ and $\mathrm{Var}_\Lambda(A) \to \|Q\hat{A}\Omega\|^2$, where Q is the projection orthogonal to the vacuum, as in the proof of [F, Theorem 'Gap $\Leftrightarrow$ clustering in time' (`thm:gapclust`)](i); and $\langle Q\hat{A}\Omega, e^{-H} Q\hat{A}\Omega \rangle \le e^{-m_{\Lambda_s}(\beta)} \|Q\hat{A}\Omega\|^2$. The bounds of Lemma 5.3 are uniform in L_0, so they hold for the limits Γ_1, Γ_0, and $e^{-m_{\Lambda_s}(\beta)} \ge \Gamma_1/\Gamma_0$.

(i) By [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) (note $\beta < \beta_0(d) < \beta_*(d)$, Lemma 4.2), the covariance and the variance under μ_β are limits of those on periodic

lattices with all sides tending to infinity, so the bounds of Lemma 5.3 hold for them. With ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)] and $\tilde{A} := A - \nu(A)$, a function of the time-zero slice, [II, Corollary `iv:cor:chain`](iv) gives $\Gamma_1 = \langle \tilde{A}, P\tilde{A}\rangle_\nu$ and $\Gamma_0 = \|\tilde{A}\|_\nu^2$; since $\tilde{A} \perp \mathbf{1}$ and $\|P|_{\mathbf{1}^\perp}\| = e^{-m(\mu_\beta)}$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)], $\Gamma_1 \le e^{-m(\mu_\beta)}\Gamma_0$.

Finally, $c(\beta)$ is bounded, $\theta(\beta)$, $x(\beta)$ and $\hat{q}(\beta)$ are $O(\beta)$, and $u(\beta) \ge (1+\iota)\beta e^{-\beta}/(2N^2)$ (Lemma 5.1(iv)), so $\delta(\beta) = O(\hat{q}^5/\beta^4) = O(\beta)$ and $D(\beta) = O(\beta)$. □

Corollary 5.5 (The gap is $4\log(1/\beta) + O(1)$)**.** *Under the hypotheses of Theorem 5.4,*

$$4\log\frac{1}{\beta} - 4\log(4e^3K) - 4\log\frac{e^{4\beta}-1}{4\beta} \;\le\; m(\mu_\beta) \;\le\; 4\log\frac{1}{\beta} + 4\log\frac{2N^2}{1+\iota} + 4\beta + D(\beta),$$

and the same bounds hold for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 4. In particular $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$, and the same holds uniformly for the finite-volume gaps on spatial lattices with all sides ≥ 4.

Proof. The lower bounds are Theorem 1.2(ii) and (i), with $q(\beta) = 4e^3K\beta \cdot \frac{e^{4\beta}-1}{4\beta}$. The upper bounds are Theorem 5.4 with $\log\frac{1}{u} \le \log\frac{2N^2}{(1+\iota)\beta} + \beta$ (Lemma 5.1(iv)). □

Remark 5.6 (The threshold of Theorem 5.4). Numerically (`python3`), $\delta(\beta) \approx 1.0 \cdot 10^{16}\beta$ for $SU(2)$ in $d = 4$, $\approx 1.0 \cdot 10^{15}\beta$ for $SU(2)$ in $d = 3$, $\approx 1.9 \cdot 10^{19}\beta$ for $SU(3)$ in $d = 4$, and $\approx 3.2{\cdot}10^{14}\beta$ for $U(1)$ in $d = 4$; the condition $\delta \le \frac{1}{2}e^{-48(d-1)\theta}$ holds for $\beta \le 4.8{\cdot}10^{-17}$, $5.0{\cdot}10^{-16}$, $2.6 \cdot 10^{-20}$ and $1.5 \cdot 10^{-15}$ respectively. These numbers have no physical meaning. They come from comparing the exact leading term, in which each of the four faces carries the dimension-reduced weight $u \approx (1+\iota)\beta/(2N^2)$, with sup-norm bounds on the remainder, in which each plaquette carries $\approx 4\beta$ times a polymer-counting factor $\approx eK$. A bound on the remainder through the character expansion, giving each face of a closed surface a weight of order u, would remove most of this loss; it is not attempted here. The same constant enters $D(\beta)$: $D(\beta)/\beta$ tends to the same limit as $\delta(\beta)/\beta$ (about $1.0 \cdot 10^{16}$ for $SU(2)$ in $d = 4$), so the $O(\beta)$ in Theorem 1.3 and Corollary 5.5 carries it; at the thresholds just quoted, $D = \log 2 + o(1)$. What the theorem establishes is the asymptotic statement of Corollary 5.5.

6 Two dimensions; discussion

Proposition 6.1 (Two dimensions)**.** *Let $d = 2$ and $0 < \beta < \beta_0(2)$, so that $K = 4$ and $q(\beta) < 0.909$ (Lemma 4.2).*

(i) For every $L_1 \ge 3$, the spatial lattice $\Lambda_s = \mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta) \ge L_1 \log(1/q(\beta))$.

(ii) In infinite volume, $\mathrm{Cov}_{\mu_\beta}(A, B) = 0$ for all A, B as in Theorem 1.2(iii) (time ranges with $a_2 < b_1$); hence $m(\mu_\beta) = \infty$.

Proof. By Lemma 3.1, $g = L_1$, and Theorem 4.1 gives the bound with $q^{L_1\tau}$. (i) follows as in the proof of Theorem 1.2(i), with L_1 in place of 4. For (ii), the bound for the periodised observables on tori with $L_0, L_1 \to \infty$ tends to 0, since $\tau \ge 1$ and $g = L_1 \to \infty$, and the torus covariances converge to $\mathrm{Cov}_{\mu_\beta}(A, B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2) < \beta_*(2)$, Lemma 4.2); then [II, Theorem `iv:thm:OS`](iii) applies with every $m > 0$. □

Remark 6.2 (Consistency with the exact solution). In $d = 2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_r c_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the exact finite-volume gap is $L_1 \log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 6.1 reproduces both features (with q in place of

u). It is also consistent with [II, Proposition 'Gauge invariance of the ground-state chain' (`iv:prop:phys`)]: in $d = 2$ the spatial links of a time slice form a line, gauge-invariant functions of finitely many of them are constant, and P acts trivially on the physical sector. The difference between $d = 2$ and $d \geq 3$ is the girth of the spatial graph (L_1 versus 4): in $d \geq 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 6.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta) \sim 4\log(1/u(\beta))$. Theorem 1.2 gives the lower half, $m \geq 4\log(1/\beta) - O(1)$ for all $\beta < \beta_0(d)$, and Theorem 5.4 the upper half, $m \leq 4\log(1/u(\beta)) + O(\beta)$ for sufficiently small β; Corollary 5.5 combines them. The upper bound uses a single correlator at time separations 0 and 1; by the positivity of P ([II, Remark 'Consequences for the numerics' (`iv:rem:numerics`)]) any larger pair of separations could be used instead. The additive constants of the two halves differ ($-4\log(4e^3K)$ versus $+4\log\frac{2N^2}{1+\iota}$); the strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$, i.e. the upper one. (ii) The constant $-4\log(4e^3K)$ reflects the crude counting of [III] and the crude bound $\|g_p\|_\infty \leq e^{4\beta} - 1$: for $SU(N)$ and $U(1)$, $c_0 = 1 + O(\beta^2)$ gives $\|g_p\|_\infty \leq 2\beta + O(\beta^2)$, which improves the constant by $4\log 2$, and counting only surfaces without free edges, as in [III, Remark 'Constants, sharpness, improvements' (`al:rem:constants`)](ii), would improve it further. (iii) The hypothesis $L_i \geq 4$ in (i) is needed: if a spatial side equals 3, Theorem 4.1 holds with $g = 3$, and closed strips of three plaquettes per layer around that side (torelons, carrying the flux of a Polyakov loop) are expected (heuristic) to give states of energy $\approx 3\log(1/u(\beta))$, below $4\log(1/q(\beta))$ for very small β. (iv) Nothing here is non-abelian: the argument applies verbatim to $G = U(1)$, whose $d = 4$ theory is expected to be massless at large β (proved in [FS82] for the Villain action and asserted there for Wilson's; [III, Remark 'The $U(1)$ test' (`al:rem:u1test`)]). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is controlled with the right leading behaviour (Corollary 5.5), which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have $\geq g$ temporal plaquettes per crossed layer ($g = 4$ for $d \geq 3$, spatial sides ≥ 4)	Lem. 3.1, 3.2, 3.3, Cor. 3.4	Theorem
Covariance bound $q^{g\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m(\mu_\beta) \geq 4\log(1/q(\beta)) = 4\log(1/\beta) - O(1)$, and $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$ for spatial sides ≥ 4; $d \geq 3$, $\beta < \beta_0(d)$	Thm. 1.2	Theorem
$d = 2$: gap $\geq L_1 \log(1/q)$; infinite gap in infinite volume	Prop. 6.1	Theorem
Leading term of the plaquette–plaquette covariance: $\tau_0 u^4(1 + O(\beta))$, the duplicated term $\Phi_F(X_0) = 2\tau_0 u^4$ being exact (four sides of a cube); unique minimal connecting set	Lem. 5.1, 5.2	Theorem
Matching upper bound $m \leq 4\log(1/u(\beta)) + O(\beta)$ (infinite volume; finite volume, sides ≥ 3), for sufficiently small β	Thm. 5.4	Theorem
$m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$ (also finite volume, sides ≥ 4)	Cor. 5.5	Theorem
Upper bound for β of order 10^{-4} (character-expansion bound for the remainder)	Rem. 5.6	open

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